Identifying the Solar Sources of Ambient Solar Wind Driven Substorms with the ADAPT-WSA Model

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~ Introduction ~

Science Objectives: To characterize the solar sources and solar wind properties of intervals in which geomagnetic substorms occur in Earth's magnetosphere, outside of times when there are large-scale solar eruptions.

Why it matters?: The effects of the large-scale solar eruptions on Earth have been well-researched. However, the constant outflow from the Sun that is the solar wind forms the ground-state of space weather as it continually bombards Earth's magnetosphere with magnetic fields and electrically charged particles. This project is a critical first step to one day being able to forecast when the ambient solar wind will drive substorms and other geomagnetic activity in Earth's magnetosphere.

***Key Words**: Geomagnetic Substorm Origins, Solar Wind*

~ Methods ~

We executed this analysis following these steps:

1. **Data Sources**:
   a. **Substorm Onset Times**: SuperMAG (Forsyth, Newell, and Ohtani datasets).
   b. **In-Situ Solar Wind Properties**: ACE spacecraft.
   c. **Model Proxies for Solar Wind Source Region**: ADAPT - WSA coronal model.
2. **Time Corrections**:
   a. Applied time-offset corrections to WSA and ACE data by calculating the solar wind travel time to Earth using t = d/v.
   b. In cases where velocity was not recorded, a default value of 400 km/s was used to maintain solar wind consistency.
3. **Data Sorting**:
   a. WSA data was filtered within ± 3 hours of each substorm onset time.
   b. ACE data was filterd within ± 1 hour of each substorm onset time.
   c. Substorm events were excluded if they occurred less than 3 days after a recorded CME event. This ensured that our study included substroms driven purely by the ambient solar wind.
4. Once sorting was completed, we were left with **123 substorms** in which we characterized their solar wind source region and in-situ solar wind properties.

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~ Results ~

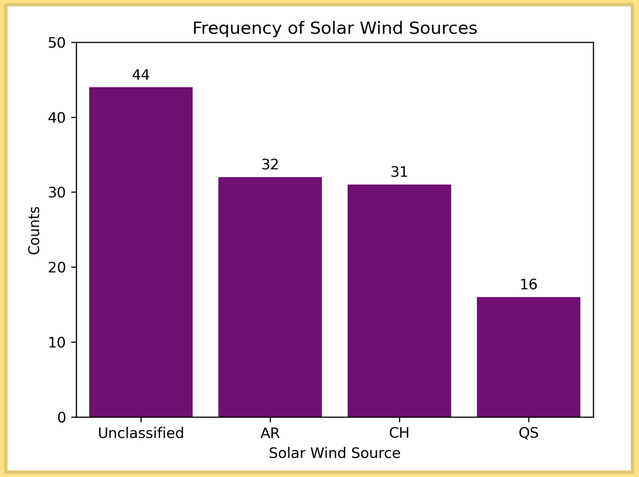

Fig. 1. Solar Wind Source v. Number of Substorms
Solar wind sources were determined by separating populations of solar wind using the model-derived coronal hole boundary distance (see below) and the magnetic field strength at each source.

- **Coronal holes (CH):** B < 20 G, DCHB > 4 (i.e. weak magnetic field, far from the coronal hole boundary)
- **Active region (AR):** B > 20 G, DCHB < 4 (i.e. strong magnetic field, near the coronal hole boundary)
- **Quiet Sun (QS):** B < 20 G, DCHB < 4 (i..e weak magnetic field, near the coronal hole boundary)
- **Unclassified:** Does not have a designated Solar Wind Source, see *Future Work*.

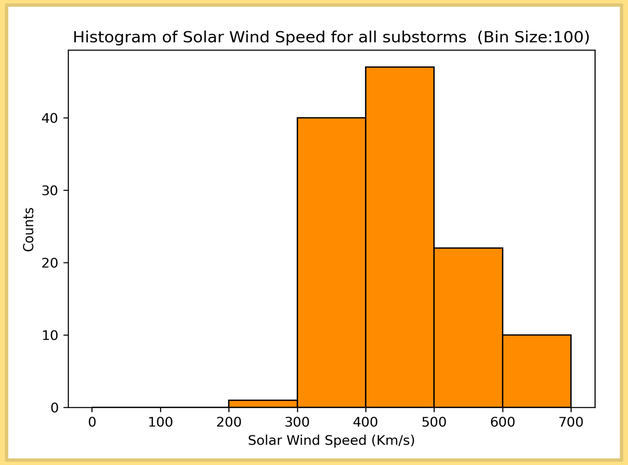

Fig. 2. Solar Wind Speed (km/s) v. Number of Substorms
~75% of these events are driven by slower solar wind (v < 600 km/s).

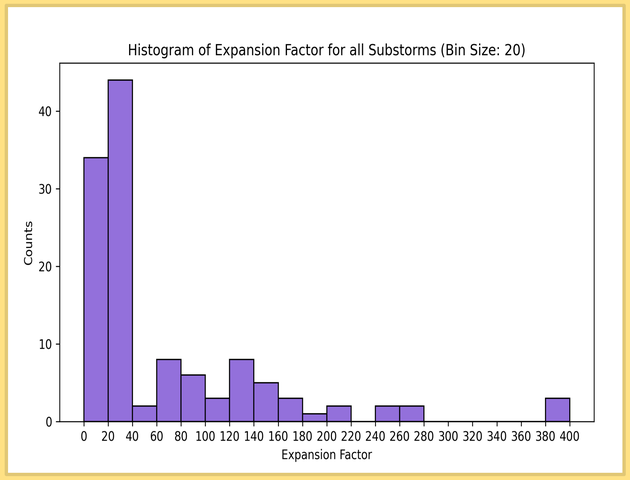

Fig. 3. Expansion Factor v. Number of Substorms
50% of these substorms originate from a solar source region with low magnetic expansion factor (fs < 40). This suggests that substorms originate from locations with a wide range of magnetic field strengths.

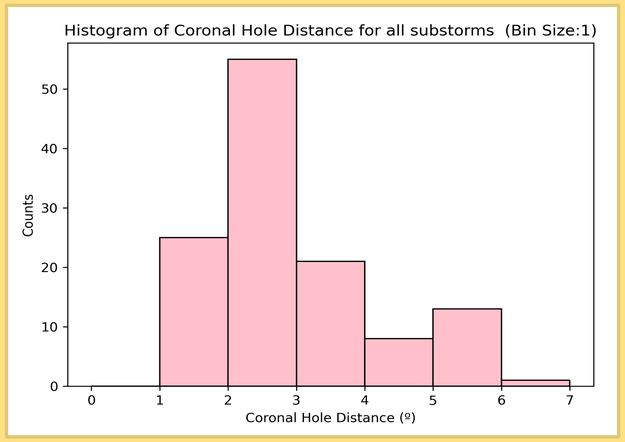

Fig. 4. Coronal Hole Distance (º) v. Number of Substorms
~ 75% of these events originate from the coronal hole boundary (DCHB < 4). Solar wind that originates from coronal holes boundaries is largely driven by inter-change reconnection.

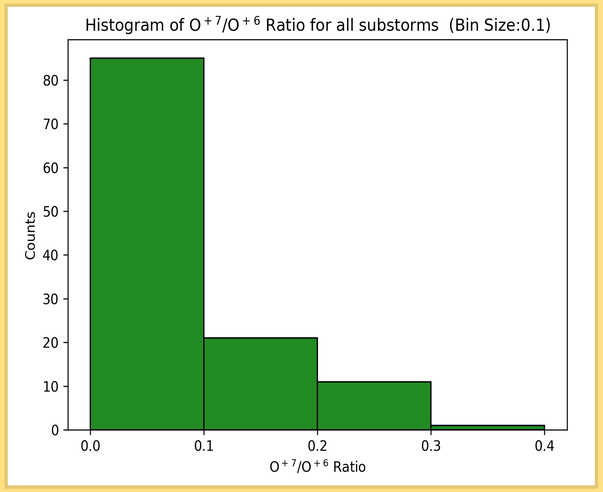

Fig. 5. O^{7+}/O^{6+} v. Number of Substorms
75% of these substorms originate from regions with low O^{7+}/O^{6+} . This is typically indicative of coronal hole origin, however, a wide range of values for Fe/O is expected at the coronal hole boundary due to the mixing of plasma from various sources.

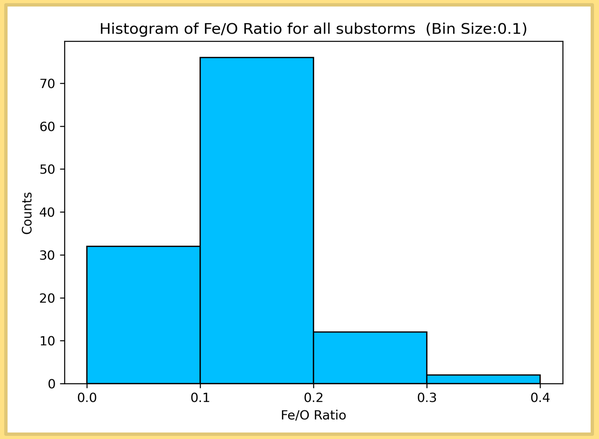

Fig. 6. Iron/Oxygen Ratio v. Number of Substorms
Over 75% of the these substorms originate with low Fe/O ratios (Fe/O < 0.2). This is typically indicative of coronal hole origin, however, a wide range of values for Fe/O is expected at the coronal hole boundary due to the mixing of plasma from various sources.

~ Summary ~

Biggest takeaways:

- Of the 5 months of data we analyzed, sampling periods of both high and low activity in the solar cycle, we find evidence that the majority of these substorms originate from coronal hole boundaries at the Sun, a highly dynamic region where where interchange reconnection is driving the formation of the solar wind. This is most supported by the results in Fig. 2 and 4.

~ Future Work ~

Next steps:

- Revisit unclassified solar wind sources that did not fit our original criteria for defining sources of the solar wind.
- Normalize results based on all possible solar wind sources during each solar rotation we investigated.
- Expand dataset to include all of solar cycle 24. This will allow us to have more conclusive results.